METHOD 24 / A COMPLETE BEGINNER GUIDE

Multifractal DCCA

How does cross-correlation of detrended FX profiles change when you weight large fluctuations more? Multifractal detrended cross-correlation analysis (MF-DCCA) extends DCCA (folder 23) with an order parameter q.

This guide uses **positive q only**. At q = 2 you recover ordinary DCCA scaling behaviour. Teaching numbers only; experiments in this folder have not started.

Read it in three parts

First: Idea, percent returns, and profiles. **Next:** Kwapie■ et al. (2015) context and the G and rho_q formulas. **Last:** A hand check, limits, and software notes.

Pages	What you will learn
2	Idea; is rho_q a percent?
3	FX percent on 1.1000 to 1.1110
4	Profiles, scales s, and segments v
5	Research: Kwapie■, O■wi■cimka, Dro■d■ 2015
6-9	Symbols, MF-DCCA formulas, hand calc
10-11	Positive q, q = 2 link to DCCA
12-14	Limits, checks, and sources

Prepared for the folder: 24 - Multifractal DCCA
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02 / THE BASIC IDEA

What does MF-DCCA do?

Start from two cumulative return profiles (integrated series). Cut them into windows of length s, remove a local trend in each window, and form a cross-fluctuation measure in segment v at scale s.

The score rho_q(s)

```
G_xy(q,s) = mean_v[ sign(c_v) * |c_v|^(q/2) ]
G_xx(q,s) and G_yy(q,s) use the same rule on each leg

rho_q(s) = G_xy(q,s) / sqrt( G_xx(q,s) * G_yy(q,s) )
```

Here c_v is the detrended cross-covariance fluctuation in segment v (notation varies in papers; see source [1]). rho_q is on roughly [-1, +1] after scaling, like a correlation. It is **not** a price percent.

Is rho_q a percent?

No. Returns may be labelled 1.00%. rho_q describes weighted co-movement of detrended fluctuations at scale s. q is a dimensionless order, not a percent.

03 / RETURNS IN PLAIN NUMBERS

FX percent math on one move

Take a quote move from **1.1000 to 1.1110** in one bar (same units).

```
Simple return = 1.1110 / 1.1000 - 1 = 0.01 (1.00% label)
Log return = ln(1.1110 / 1.1000) = about 0.009950
```

MF-DCCA is built on return series (often log returns), then cumulative profiles. Use one return definition for both legs. Document quote direction and bar time before you integrate.

Integrating returns builds the profile; MF-DCCA then studies how cross-fluctuations scale with window length s. A 1% daily move is input data, not the output rho_q.

04 / DATA RULES

Profiles, scales, and segments

Same length, aligned time

Both profiles must share the same index after integration. Missing bars break segment counts; record your gap rule in experiment notes when v001 exists.

Scale s

s is the window length in points (bars). You compute rho_q(s) for several s values and study scaling. This guide shows one scale with three teaching segments v = 1, 2, 3.

Relation to folder 23

Standard DCCA is the q = 2 case of the same G construction. Multifractal analysis asks whether rho_q(s) changes with q, which signals heterogeneous fluctuation sizes across segments.

We do not invent EURUSD scaling exponents here. When experiments start, tie every plot to a snapshot id and fixed (q, s) grid.

05 / RESEARCH NOTES

Where did MF-DCCA come from?

Kwapie■, O■wi■cimka, Dro■d■ (2015) [1]

Physica A, 436, 122-132. The paper develops MF-DCCA for multifractal cross-correlations between two time series, extending the DCCA family with the order q generalisation used in this guide.

Earlier DCCA work (folder 23) removed local trends before measuring cross-correlations at long horizons. MF-DCCA adds the q-weighted average so large and small fluctuations can contribute differently.

What this guide claims

We teach the rho_q scaling definition and the q = 2 link to DCCA. Full scaling regressions, spectrum estimation, and bootstrap inference belong in the paper and in your analysis code.

Check DOI 10.1016/j.physa.2015.05.028 when you need exact algorithm steps and figure notation.

06 / DICTIONARY

Meet the symbols

Symbol	Meaning
q	Order parameter; q > 0 in this guide; q = 2 is DCCA.
s	Segment (window) length in points.
v	Segment index after tiling the profile.
c_v	Detrended cross-fluctuation in segment v at scale s.
G_xy(q,s)	q-weighted mean of signed power terms across v.
rho_q(s)	Normalised cross measure at order q and scale s.

G_xx and G_yy apply the same mean_v[sign(c)|c|^(q/2)] pattern to each single-series fluctuation series inside segment v.

Software packages may name c_v differently; map their output to G_xy before you compare to rho_q here.

07 / WHAT EACH PART DOES

The main formulas

```
Term_v(q) = sign(c_v) * |c_v|^(q/2)

G_xy(q,s) = (1/V) * SUM_v Term_v(q) on cross c_v

rho_q(s) = G_xy / sqrt(G_xx * G_yy)
```

When q = 2

Then |c_v|^(q/2) = |c_v| and sign(c_v)*|c_v| = c_v. The cross term becomes an ordinary average of c_v across segments, matching the DCCA-type q = 2 case described in [1].

Positive q only here

Negative q emphasises small fluctuations and needs extra care with zeros and estimation noise. This beginner guide stays with q > 0 so terms stay real for nonzero c_v.

rho_q is not a return forecast. It summarises how detrended cross-fluctuations co-scale at s for the chosen q.

Calculate by hand: q = 2

Three teaching segments at one scale s. Cross-fluctuations c_v (decimals) and single-leg fluctuation magnitudes for G_xx and G_yy:

v	c_v (cross)	f_x,v	f_y,v
1	0.0002	0.0010	0.0011
2	0.0003	0.0020	0.0019
3	0.0001	0.0015	0.0014

For q = 2, each G is the plain mean of the column (because sign(c)*|c| = c for cross, and f columns are positive teaching magnitudes).

```
G_xy(2,s) = (0.0002 + 0.0003 + 0.0001) / 3 = 0.0002
G_xx(2,s) = 0.0015
G_yy(2,s) = (0.0011 + 0.0019 + 0.0014) / 3 = about 0.001467

rho_2(s) = 0.0002 / sqrt(0.0015 * 0.001467)
= about 0.1348
```

This tiny table only checks algebra. Real MF-DCCA also loops over many s and may estimate multifractal spectra from how G behaves with s.

09 / READING Q

Changing q changes emphasis

Larger q

Larger q puts more weight on segments where |c_v| is big. rho_q(s) can differ from rho_2(s) when large and small co-fluctuations disagree.

Do not mix with Pearson on returns

Same-bar Pearson (folder 01) does not detrend local segments or scan scales. MF-DCCA answers a long-memory / scaling question, not a one-bar linear tilt.

When you report rho_q, always give q, s, return type, and segment construction. A single number without scale context is easy to misread.

10 / PRACTICE HABITS

Using MF-DCCA on FX

Long samples

You need enough points to tile multiple windows at each s. Very short histories make rho_q(s) unstable.

Sessions and microstructure

Changing bar size changes s in physical time. A 5-minute segment is not the same object as a daily segment.

No invented pair results

Experiments are not started. Do not cite made-up EURUSD multifractal spectra in reports. Future work uses frozen snapshots per docs/VERSIONING.md.

Limits and common errors

Not a trading signal

Scaling similarity does not, by itself, imply profitable hedges or lead-lag rules.

Trend removal choice

Polynomial detrending degree and overlapping vs non-overlapping windows change c_v. Match your software to the paper you cite.

Versus DCCA (folder 23)

Folder 23 is the q = 2 detrended cross-correlation path. Folder 24 adds the q spectrum for heterogeneous fluctuation sizes.

Versus Pearson rolling (folder 05)

Rolling Pearson tracks local linear correlation. MF-DCCA studies detrended fluctuation scaling across window lengths.

Check your understanding

A. Is rho_q a percent?

No. Returns can be percent-labelled; rho_q is a scaled cross-fluctuation score.

B. What does q = 2 mean?

You recover the DCCA-type averaging of c_v in the G_xy construction used here.

C. Can I use negative q in this guide?

Not in this beginner path; positive q only. See [1] for the full multifractal treatment.

D. One scale, three segments

The hand example checks rho_2(s) algebra. Full studies loop s and q and assess uncertainty.

Save segment c_v lists from code and reproduce G_xy and rho_q in a spreadsheet when experiments begin.

Calculate with software

```
from math import sqrt

def g_q(vals, q):
return sum((1 if c>=0 else -1)*abs(c)**(q/2) for c in vals)/len(vals)

cross = [2e-4, 3e-4, 1e-4]
fx = [1e-3, 2e-3, 1.5e-3]
fy = [1.1e-3, 1.9e-3, 1.4e-3]
rho2 = g_q(cross,2)/sqrt(g_q(fx,2)*g_q(fy,2))
print(rho2) # about 0.1348
```

Implement full profile integration, tiling, and detrending before you apply this to snapshot data. Compare one (q,s) row to hand arithmetic first.

Sources and further reading

Links checked on 3 November 2026. Hand arithmetic is original teaching material. Bracket numbers point to primary references.

[1] Kwapie■, J., O■wi■cimka, P., & Dro■d■, S. (2015). Multifractal cross-correlations between the world oil and precious metal markets. Physica A, 436, 122-132.
MF-DCCA definition and q order; context for G and rho_q.
Open source

[2] Folder 23 DCCA detrended cross-correlation README.
q = 2 baseline method in this project tree.
Open source

Next step: experiments/v001 with logged (q, s) grids and detrending settings copied from your chosen library.

Workflow before experiments

1. Write the question first

Example: At daily scale, do EURUSD and USDJPY detrended cross-fluctuations show q-dependent rho_q(s) over s = 32..256? Name returns, snapshot id, and segment rule.

2. Build profiles once

Integrate log returns from the same aligned calendar. Save the integrated series and the return type in experiment notes.

3. Fix q and s grids before you look

Pre-specify positive q values and window lengths s. Scanning many pairs without correction inflates false findings.

4. Report rho_q with context

Give q, s, sample length, detrending degree, and segment count V. State that rho_q is not a price percent and not a causal forecast.

When v001 exists, attach one CSV of c_v values and a screenshot of rho_q(s) versus s for a single q to show reproducibility.